Cross-Scale Mixing Attention for Multisource Remote Sensing Data Fusion and Classification

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***Abstract*— Hyperspectral and multispectral images (HS/MS) fusion and classification as an important branch of data quality improvement and interpretation have attracted increasing attention in recent years. However, the unavailable sensor prior still limits the performance of many traditional fusion methods, consequently deteriorating the classification results. Despite the unsupervised methods based on convolutional neural network (CNN) making a lot of attempts to mitigate the limitations, challenges with extracting the long-range dependencies hamper the performance. To address these impediments, a transformer-based baseline constructed by the cross-scale mixing attention transformer (CSMFormer) is designed for HS/MS fusion and classification. Especially, the spatial–spectral mixer (SSMixer) is utilized to extract the long-range dependencies at a large scale. Simultaneously, cross-scale feature calibration is achieved by combining information from the original scale. After that, the nonlinear enhancement module (NLEM) is designed to encourage feature discrimination. Note that the spatial and spectral mixers can be replaced by any spatial–spectral feature extractors. Therefore, the proposed CSMFormer is flexible in data fusion, land-covers' classification, segmentation, and so on. Experiments about data fusion and land-covers' classification on two HS/MS wetland remote sensing scenes demonstrate the superiority of the proposed CSMFormer baseline, improving the data quality and classification precision.**

***Index Terms*— Cross-scale mixing attention transformer (CSMFormer), data fusion, hyperspectral and multispectral images (HS/MS), land-covers' classification, long-range dependencies.**

I. Introduction

HYPERSPECTRAL imaging technology collects the reflecting optical intensity of objects in the visible and infrared spectral range [1], [2]. Therefore, hyperspectral image (HSI) is particularly useful for land-covers' classification in wetland scenes by analyzing the discriminative spectrums. However, hyperspectral imaging relies on long-time exposure to obtain a sufficient signal-to-noise ratio, which also leads to relatively low spatial resolution. In contrast, multispectral image (MSI) has the advantage of high spatial resolution and is complementary to HSI [3]. Therefore, hyperspectral and multispectral images (HS/MS) fusion is able to generate high spatial–spectral resolution images (HR-HSI) by absorbing the complementary merits from low-resolution HSI (LR-HSI) and high-resolution MSI (HR-MSI), and then, an elegant classifier is employed for land-covers' classification.

To establish an effective HS/MS fusion model, researchers have carried out a lot of algorithms that are divided into two categories. One is that traditional methods include matrix-based methods, Bayesian-based methods, and tensor-based methods. In [4], the nonparametric Bayesian strategy was further employed to generate the sparse code. In [5], a coupled matrix factorization problem with constraints from spectral mixing was solved to reconstruct HR-HSI. Although matrix factorization and low rank are the popular schemes for HSI super-resolution [6], some limits are still exposed. However, the matrix-based methods inevitably unfold the 3-D HSI as a 2-D matrix, in which the inherent spatial–spectral structures are ignored. The tensor-based methods have attracted more attention. In [7], the nonlocal sparse tensor factorization was utilized for HSI super-resolution. In [8], the problem of image fusion was redefined as the estimation of core tensor and dictionaries of the three modes, which were achieved by the coupled tensor factorization of LR-HSI and HR-MSI. In [9], the complementary merits for data fusion were achieved by the spatial consistency and spectral smoothness, which were extracted from the LR-HSI and HR-MSI, respectively. Xu et al. [10] designed a high-order coupled tensor ring (TR) to represent the spatial–spectral structures, and then, the HR-HSI was reconstructed by the coupled TR representation. In addition, patch clustering and factorization models have been successfully adopted in HR-HSI reconstruction [11], [12], [13].

The other one is a fusion model based on deep learning. Deep networks naturally learn semantic features from low level to high level in an end-to-end manner. Especially, the hierarchical representation makes the convolutional neural network (CNN) powerful in visual representation [14], [15]. Over the past years, CNN has shown great potential in the field of data fusion. Since Palsson et al. [16] adopted 3-D-CNN for HS/MS fusion, numerous CNN-based models have been successfully proposed. Xie et al. [17] developed an MS/HS fusion model based on proximal gradient, in which the proximal operators and model parameters were optimized by CNN. Dian et al. [18] employed the low-dimensional subspace multiplied by the coefficients to estimate the HR-HSI, and then, the well-trained CNN denoiser was employed to regularize the estimation of coefficients. Xu et al. [19] designed a multiscale network with the root mean square error (RMSE), angle, and Laplacian loss for HS/MS fusion.

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Zhang et al. [20] developed the spatial–spectral reconstruction network (SSRNet) for HS/MS fusion. Gao et al. [21] fused the LR-HSI and HR-MSI via a CNN, using spatial–spectral vision transformer (SSViT) for classification. Li et al. [22] presented an unsupervised network to reconstruct the desired HR-HSI. Li et al. [23] proposed a HyperNet for HS/MS and panchromatic image fusion.

HS/MS fusion in practice usually serves as a preprocessing step for downstream vision tasks, such as classification [24], [25] and detection [26]. Thus, the classification performance on HR-HSI can be a task-specific evaluation criterion. Recently, many researchers have been dedicated to developing discriminative features [27], [28], and then, powerful classifiers are employed for land-covers' classification [29]. In the past decade, classification technologies, such as support vector machine (SVM) [30], random forest (RF) [31], Gaussian process [32], and extreme learning machine (ELM) [33], have been brilliant in HSI classification. Zhao et al. [34] employed RF into a stacked sparse autoencoder to improve the generalization performance. Li et al. [35] adopted the deep belief network (DBN) for HSI classification. Zhong et al. [36] improved the DBN with a diversity-promoting prior. However, the above technologies forcibly convert HSI cubes (3-D) into patches (2-D) to adapt to the input requirements. Thus, the classification performance deteriorates due to the loss of spatial information.

To further address these limitations, the CNN-based method allows 3-D cubes as input, providing a natural way to learn the discriminative features for classification [37], [38]. Chen et al. [39] were among the first to use CNN to classify HSI. Since then, HSI classification has been explored. In [40], the pyramidal residual network was designed to extract the multiscale spatial–spectral features. In [41], the capsule network was employed for HSI classification, which significantly reduced the complexity of the network designer. In [42], discriminative spatial–spectral features were captured by the attention network. Cao et al. [43] developed an active deep learning method for HSI classification, which significantly improved the performance with few labels. Gao et al. [44] joined feature space and label space constraints to enhance image feature discrimination. Liu et al. [45] designed a central attention network for HSI classification. Li et al. [46] utilized the synthetic aperture radar (SAR) images to assist HSI classification. Some equally important work employed recurrent neural networks (RNNs) for land-covers' classification, such as [47], [48], and [49].

In addition to the CNN-based method, vision transformer (ViT) and its extensions (as shown in Fig. 1) have also boosted the performance of a series of remote sensing tasks, such as data fusion and classification. Specifically, the research about MetaFormer [50] shows that the elegant structure of ViT contributes to the improvements, including the token mixer and channel multilayer perceptron (MLP), which is utilized to encode long-range dependencies and channel interaction. For example, as shown in Fig. 1(a) the token mixer of ViT is achieved by self-attention, and LN represents the layer normalization. More recently, ViT-based methods have attracted a lot of attention in the remote sensing community. Hong et al. [51] developed a novel spectral transformer (SpecFormer) to extract the spectral correlation. He et al. [52] proposed a transformer-induced hierarchical graph network for multimodal semantic segmentation in remote sensing scenes. In [53], the ViT was employed to guide small networks to image scene classification. In [54], a transformer-based enhancement network is designed for reconstruction by emphasizing high-dimensional feature representation.

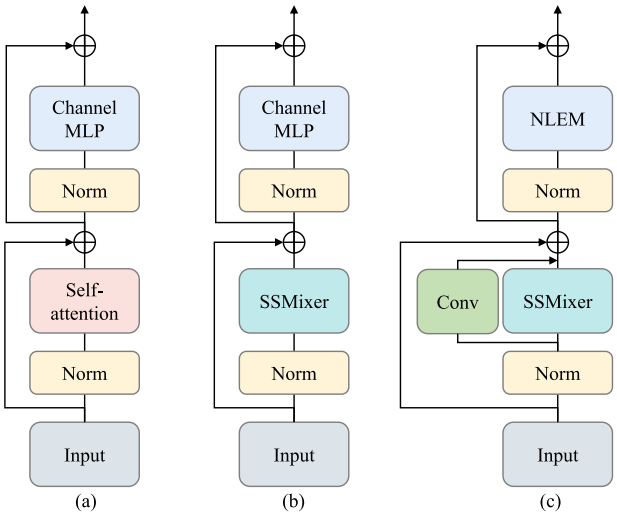


Fig. 1. Illustration of different Former-based methods. (a) ViT. (b) LSFormer. (c) CSMSFormer.

Despite extensive research work conducted in the HS/MS fusion and classification, the high dimensionality, together with the underlying redundancy, still makes the fusion or classification problem a very challenging one. Basically, there are two common aspects that HS/MS fusion and classification need to deal with.

1) *Long-Range Spatial–Spectral Dependencies:* Using ViT as shown in Fig. 1(a) to encode long-range dependencies has proved to be effective. However, the self-attention of ViT forces the 2-D vector to the 1-D vector, resulting in the loss of spatial structure. Besides, the convolutional ViT inevitably leads to an exponential increase in the computational burden. Therefore, it is crucial for building a flexible encoder to adaptively pay close attention to long-range dependencies.
2) *Cross-Scale Feature Calibration:* As shown in Fig. 1(b), the large-scale transformer (LSFormer) extracts the long-range dependencies by spatial–spectral mixer (SSMixer). However, the contribution of original scale information is ignored when capturing large-scale long-range dependencies. Eventually, this fact leads to low feature discrimination. Therefore, it is essential to consider the information from the original scale to the large scale and adaptively reweigh their contributions.

To tackle the above issues, a transformer network with a cross-scale mixing attention transformer (CSMFormer) is proposed for HS/MS fusion and classification, denoted as CSMFormer-F and CSMFormer-C, respectively. Specifically, the baseline consists of several cross-scale mixing attention (CSMA) layers, as shown in Fig. 1(c), including SSMixer and the nonlinear enhancement module (NLEM). In the SSMixer, a large-scale branch and an original-scale branch are utilized for long-range spatial–spectral dependencies and local dependencies, respectively. Specifically, in the large-scale branch, the spatial mixer is achieved by 2-D convolution, while the

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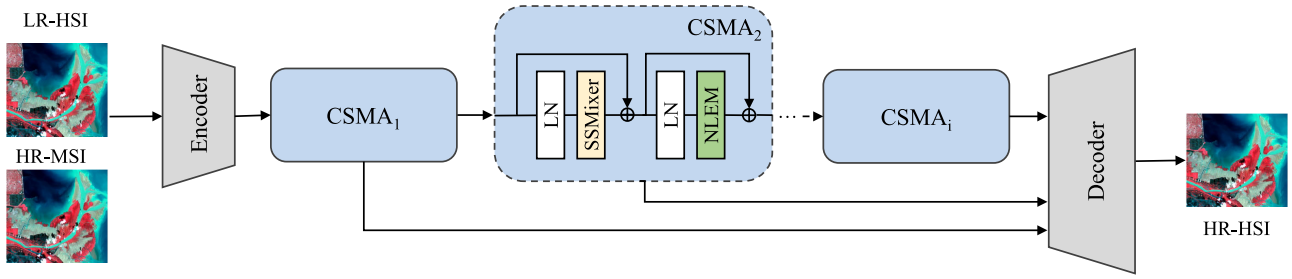


Fig. 2. Framework of the proposed CSMFormer-F for HSI and MSI remote sensing data fusion, including the encoder, CSMA layers, and decoder. The CSMA layer is composed of SSMixer and NLEM, which are, respectively, utilized for long-range spatial–spectral dependencies extraction and feature enhancement.

spectral mixer is achieved by a pointwise convolution. In the original-scale branch, local dependencies are employed for feature calibration through a pointwise convolution. Next, the NLEM is utilized to improve the feature discrimination of the cross-scale features. Our experimental results, conducted with two HS/MS datasets, demonstrate that the proposed CSMFormer baseline offers flexibility when fusing and classifying HSI and MSI, and achieves superior performance advantages over the other relevant methods.

The main contributions of the proposed CSMFormer baseline are summarized as follows.

1) A common CSMFormer baseline is developed for HS/MS fusion classification, which is able to effectively mine the long-range and local dependencies between different scales, encouraging feature discrimination. Specifically, the SSMixer is utilized to model the long-range dependencies, and the NLEM is designed for nonlinear representation. The proposed CSMFormer baseline can adapt to different downstream tasks, such as data fusion and classification by designing appropriate SSMixers.
2) For the HS/MS fusion framework, a decomposable large kernel 2-D convolution (LKC) is employed to model long-range spatial dependencies, followed by a pointwise convolution to model spectral dependencies. The LKC greatly reduces computational complexity. Therefore, compared with the self-attention that is more suitable for 1-D vector relations, SSMixer can fully mine 2-D long-range spatial–spectral dependencies with a less computational burden. After that, an NLEM is further designed for achieving the nonlinear representation.
3) For land-covers' classification, the CSMFormer-C is applied for land-covers' recognition. Empirically, LKC is not required for land-covers' classification, which greatly reduces feature discrimination. Therefore, the LKC is replaced by a multiscale strip convolution to extract long-range spatial dependencies. NLEM is also utilized for feature enhancement.

The remainder of this article is organized as follows. The proposed data fusion framework is described in Section II-A, and the classification framework is described in Section II-B. Section III validates the performance of the proposed baseline by providing a comparison with related works. Finally, Section IV concludes this article with some remarks and directs future research lines.

II. Proposed HS/MS Fusion and Classification Framework

For HS/MS fusion, the reconstructed HR-HSI $\mathbf{Y}_{\text{hsi}} \in \mathbb{R}^{H\times W\times C}$ is obtained by inputting the LR-HSI $\mathbf{X}_{\text{hsi}} \in \mathbb{R}^{h\times w\times C}$ and HR-MSI $\mathbf{X}_{\text{msi}} \in \mathbb{R}^{H\times W\times c}$, and the reference HR-HSI $\mathbf{R}_{\text{hsi}} \in \mathbb{R}^{H\times W\times C}$ is employed to supervise the model optimization. Here, H, W, and C ($h < H$, $w < W$, and $c < C$) are the height, width, and number of bands, respectively.

For classification, the patches centered at each pixel of the reconstructed HR-HSI are denoted as $\mathbf{x} \in \mathbb{R}^{p\times p\times C}$, where p is the patch size. The labels of land-covers' are predicted by the classification model.

A. CSMFormer-F for Data Fusion

The complementary merits of LR-HSI and HR-MSI are integrated to generate the HR-HSI. As illustrated in Fig. 2, the proposed CSMFormer-F for data fusion consists of an encoder, several CSMA layers, and a decoder.

1) Encoder Network: Consider that the reference image is unavailable in the real wetland scenes. For unsupervised model, the degraded HSI and MSI for training are generated by resampling the original LR-HSI, while the original LR-HSI is employed as the reference image. The resampling method is denoted as

$$\begin{aligned}\mathbf{X}^d_{\text{hsi}} &= \text{Down}(\text{Gaussian}(\mathbf{X}_{\text{hsi}}), 1/r)\\ \mathbf{X}^d_{\text{msi}} &= \mathbf{X}_{\text{hsi}}[b]\end{aligned} \tag{1}$$

where $\mathbf{X}^d_{\text{hsi}}$ and $\mathbf{X}^d_{\text{msi}}$ represent the degraded HSI and MSI for training. Down$(\cdot)$ is the downsampling operator, and r is the resolution ratio of HR-MSI and LR-HSI. Gaussian$(\cdot)$ denotes the Gaussian blur with kernel size $[5, 5]$. $b = \text{Rounding}(C_1/c)\times\omega$, and $\omega = [0, 1, \ldots, c-1]$ is the spectral index, which is utilized to select the bands from original LR-HSI $\mathbf{X}_{\text{hsi}}$. Here, $C_1 = C - C_2$ is the bands' number of visible to near-infrared in $\mathbf{X}_{\text{hsi}}$. Rounding$(\cdot)$ is the operation of rounding to 0. Therefore, the training set [LR-HSI, HR-MSI, HR-HSI] is denoted as $[\mathbf{X}^d_{\text{hsi}}, \mathbf{X}^d_{\text{msi}}, \mathbf{X}_{\text{hsi}}]$. The testing set is $[\mathbf{X}_{\text{hsi}}, \mathbf{X}_{\text{msi}}, -]$.

First, the degraded HSI and MSI are preliminarily fused for integration, as shown in Fig. 3, and then, the encoder network is employed for feature embedding. The preintegrated operator is denoted as

$$\mathbf{X}_{\text{pre}}(i) = \begin{cases}\mathbf{X}^d_{\text{msi}}[i], & \text{if } i = b\\ \text{Up}\left(\mathbf{X}^d_{\text{hsi}}[i], r\right), & \text{otherwise}\end{cases} \tag{2}$$

where $i = [1, 2, \ldots, C]$ is the number of HSI bands. Up$(\cdot)$ is the upsampling operator. As shown in Fig. 3, the bands of $\mathbf{X}^d_{\text{msi}}$ replaces the corresponding bands in $\mathbf{X}^d_{\text{hsi}}$. After that, $\mathbf{X}_{\text{pre}}$ is fed into the encoder network as

$$\mathbf{X}_e = \text{conv}_{3\times 3}\left(\mathbf{X}_{\text{pre}}\right) \tag{3}$$

where $\text{conv}_{3\times 3}$ is the convolutional layer with kernel size 3×3. $\mathbf{X}_e$ is the output of encoder network.

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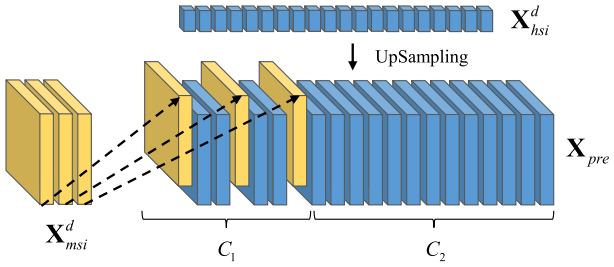


Fig. 3. Illustration of the preintegrated operator.

2) Cross-Scale Mixing Attention: To extract the long-range spatial–spectral dependencies, a novel former-based layer names CSMA is designed. First, the feature embedding from encoder network $\mathbf{X}_e \in \mathbb{R}^{H\times W\times D}$ is split into two parts along the channel dimension as

$$\begin{aligned}\mathbf{X}_e^1 &= \text{norm}(\mathbf{X}_e)[:,:,1:\eta D]\\ \mathbf{X}_e^2 &= \text{norm}(\mathbf{X}_e)[:,:,\eta D:D]\end{aligned} \tag{4}$$

where norm$(\cdot)$ is the layer normalization (LN). $\eta = 0.5$ is the split ratio. One part is fed into the attention layer to generate long-range dependencies on the large scale, and the other part is convoluted to achieve the feature calibration on the original scale. Finally, the cross-scale mixing is achieved through information propagation between the large scale and the original scale.

Different from the traditional self-attention with 1-D vector inputs, which ignores potential spatial relationships, for the large-scale branch, a 2-D LKC is necessary for capturing spatial dependencies under the large receptive field, as shown in Fig. 4. $\text{Conv}_{1\times1}$ is the pointwise convolution to aggregate the channel information. The spatial mixer and the spectral mixer are utilized to capture the spatial context and inter-channel information, respectively. Concretely, the pointwise convolution first performed for feature aggregation as

$$\mathbf{F}_e^1 = \text{Conv}_{1\times1}\left(\mathbf{X}_e^1\right) \tag{5}$$

where $\mathbf{F}_e^1$ is the aggregated feature.

Consider that the LKC is critical in data fusion, but the computing burden is heavy. Therefore, the spatial mixer is decomposed as shown in Fig. 4

$$\mathbf{Att}_s = \text{Conv}_{1\times1}\left(\text{DWConv}^d_{7\times7}\left(\text{DWConv}_{5\times5}\left(\mathbf{F}_e^1\right)\right)\right) \tag{6}$$

where $\text{Conv}_{1\times1}$ is the pointwise convolution, $\text{DWConv}_{5\times5}$ is the 5×5 depthwise convolution, $\text{DWConv}^3_{7\times7}$ is the depthwise convolution with kernel size 7×7, and the dilation rate is 3. Therefore, the above LKC decomposition is equivalent to the standard convolution kernel of 19×19. Specifically, assuming that the number of channels and convolutional kernels is c, the parameter quantity of a 19×19 standard convolution kernel is $19\times19\times c^2 = 361c^2$. As shown in (6), the decomposed LKC contains $c^2 + 7\times7\times c + 5\times5\times c = (74+c)c$ parameters. Therefore, it is found that the parameter quantity of decomposed LKC is greatly reduced.

The spectral mixer is utilized for spectral dependencies modeling by a pointwise convolutional layer as

$$\mathbf{Att} = \text{Conv}_{1\times1}(\mathbf{Att}_s). \tag{7}$$

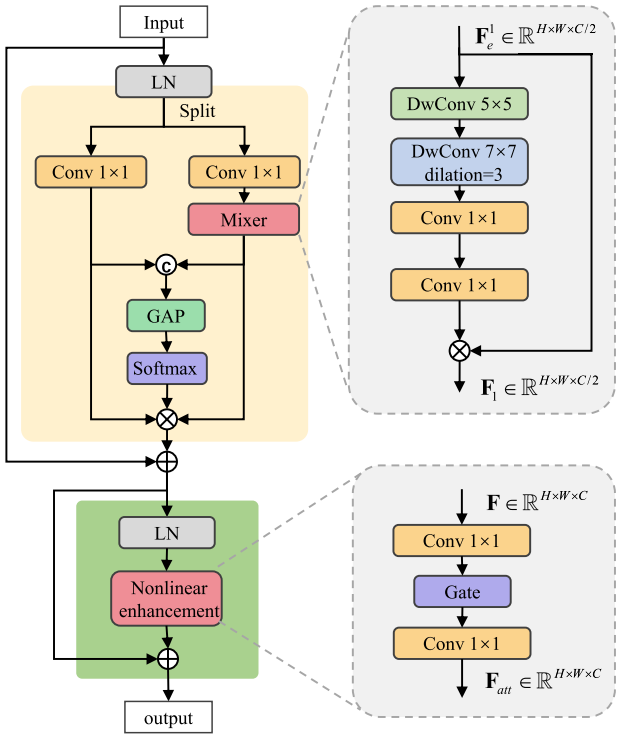


Fig. 4. Illustration of the CSMA layer for CSMFormer-F. The decomposed large kernel convolution and pointwise convolution are utilized to achieve spatial mixer and spectral mixer, respectively.

Thus, the output of the large-scale branch is calculated as

$$\mathbf{F}_1 = \mathbf{F}_e^1 \times \mathbf{Att}. \tag{8}$$

For the original-scale branch, a pointwise convolution layer is employed to obtain the feature as

$$\mathbf{F}_2 = \text{Conv}_{1\times1}\left(\mathbf{F}_e^2\right) \tag{9}$$

and then, the output of CSMA layer $\mathbf{F}$ is calculated by adaptively reweigh the contribution of $\mathbf{F}_1$ and $\mathbf{F}_2$ as

$$\mathbf{F} = \text{cat}(\omega_1\cdot\mathbf{F}_1, \omega_2\cdot\mathbf{F}_2) \tag{10}$$

where cat$(\cdot)$ is the concatenation operator along the channel dimension. ω_1 and ω_2 are the weight vectors, which are defined as

$$\omega = \sigma\left(\text{GAP}\left(\text{cat}^*(\mathbf{F}_1,\mathbf{F}_2)\right)\right) \tag{11}$$

where cat$^*(\cdot)$ is the concatenation operator along the height dimension, GAP$(\cdot)$ is global average pooling, and σ is the *softmax* function. Finally, $\omega_1 = \omega[:,0,:]$ and $\omega_2 = \omega[:,1,:]$ are employed to reweigh the contribution of different scales.

Second, the NLEM is designed to enhance feature discrimination. Consider that nonlinear functions, such as *ReLU*, *GeLU*, and *sigmoid*, improve the ability of feature representation ability. However, the nonlinear transformation inevitably leads to information loss, which is intuitively contrary to the requirements of some tasks, such as data fusion. Therefore, the Gate$(\cdot)$ function is designed to improve the flexibility of tasks while taking into account the feature representation. The NLEM is defined as

$$\mathbf{F}_{\text{att}} = \text{Conv}_{1\times1}(\text{Gate}(\text{Conv}_{1\times1}(\text{norm}(\mathbf{F})))) \tag{12}$$

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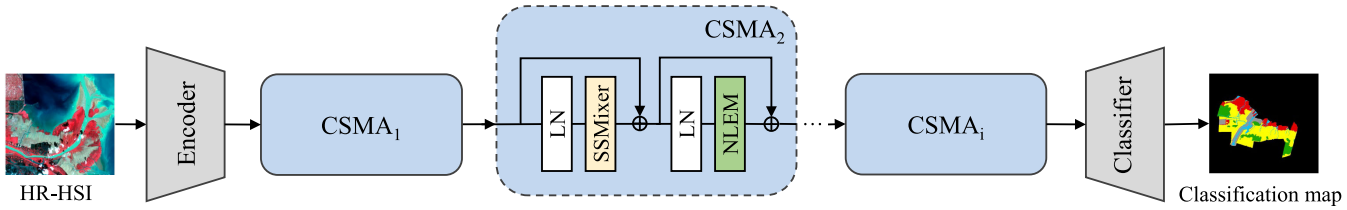


Fig. 5. Framework of the proposed CSMFormer-C for land-covers' classification, including the encoder, CSMA layers, and classifier. The CSMA layer is composed of SSMixer and NLEM, which are, respectively, utilized for long-range spatial–spectral dependencies extraction and feature enhancement.

where Gate$(\cdot)$ is the nonlinear layer, which is realized by multiplication. The nonlinear layer is defined as

$$\text{Gate}(\mathbf{Z}) = \mathbf{Z}[:,:,1:C/2] \times \mathbf{Z}[:,:,C/2:C] \tag{13}$$

where $\mathbf{Z} \in \mathbb{R}^{H\times W\times C}$ denotes the input feature.

3) Decoder Network: A decoder network is designed to reconstruct the HR-HSI. The decoder network is calculated as follows:

$$\mathbf{Y} = \text{Conv}_d(\mathbf{F}_d) \tag{14}$$

where

$$\mathbf{F}_d = \mathbf{F}_e + \mathbf{F}_{\text{att}}^1 + \cdots + \mathbf{F}_{\text{att}}^l \tag{15}$$

where Conv_d denotes that each of the three ReLU layers is followed by a convolution layer. $\mathbf{F}_d$ is the integrated feature for encoder and several CSMA layers. $\mathbf{F}_{\text{att}}^l$ is the output of the lth CSMA layer.

4) Loss Function: The data fusion model is optimized in an end-to-end manner. The loss functions are equipped to generate promising HR-HSI, including reconstruction loss, gradient loss, and spatial consistency Loss. The reconstruction loss is defined as

$$L_{\text{re}} = \frac{1}{HWC}\sum_{h=1}^{H}\sum_{w=1}^{W}\sum_{c=1}^{C}\|\mathbf{Y}(h,w,c) - \mathbf{X}_{\text{hsi}}(h,w,c)\|_2 \tag{16}$$

where $\mathbf{Y}(h,w,c)$ is the reconstructed HR-HSI. $\|\cdot\|_2$ is the L2-norm.

To ensure the texture similarity between the reconstructed HR-HSI and the reference image, the horizontal and vertical gradient losses are utilized for constraint. The gradient loss function is defined as

$$L_{\text{spa}}^h = \frac{1}{(H-1)WC}\sum_{h=1}^{H-1}\sum_{w=1}^{W}\sum_{c=1}^{C}\left\|\hat{\mathbf{E}}_h(h,w,c) - \mathbf{E}_h(h,w,c)\right\|_2$$
$$L_{\text{spa}}^w = \frac{1}{H(W-1)C}\sum_{h=1}^{H}\sum_{w=1}^{W-1}\sum_{c=1}^{C}\left\|\hat{\mathbf{E}}_w(h,w,c) - \mathbf{E}_w(h,w,c)\right\|_2 \tag{17}$$

where

$$\hat{\mathbf{E}}_h(h,w,c) = \mathbf{Y}(h+1,w,c) - \mathbf{Y}(h,w,c)$$
$$\mathbf{E}_h(h,w,c) = \mathbf{X}_{\text{hsi}}(h+1,w,c) - \mathbf{X}_{\text{hsi}}Y(h,w,c) \tag{18}$$
$$\hat{\mathbf{E}}_w(h,w,c) = \mathbf{Y}[h,w+1,c] - \mathbf{Y}[h,w,c]$$
$$\mathbf{E}_w(h,w,c) = \mathbf{X}_{\text{hsi}}[h,w+1,c] - \mathbf{X}_{\text{hsi}}Y[h,w,c]. \tag{19}$$

Thus, the gradient loss is defined as

$$L_{\text{spa}} = 0.5\cdot L_{\text{spa}}^h + 0.5\cdot L_{\text{spa}}^w. \tag{20}$$

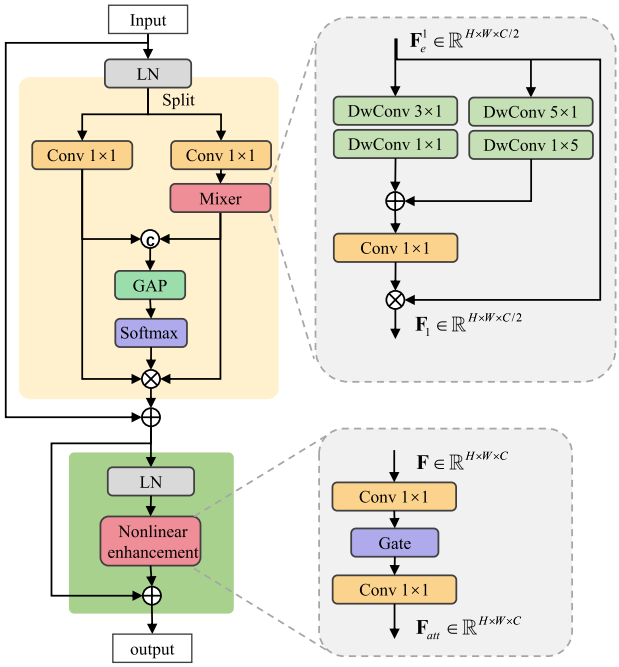


Fig. 6. Illustration of the CSMA layer for CSMFormer-C. The multiscale strip convolution and pointwise convolution are utilized to achieve spatial mixer and spectral mixer, respectively.

To enhance the spatial coherence of the enhanced image, the spatial consistency loss [55] is utilized to preserve the difference of neighboring regions (contrast ratio). The spatial consistency loss is defined as

$$L_{\text{con}} = \frac{1}{R}\sum_{i=1}^{R}\sum_{j\in\Omega(i)}\left(\left|\mathbf{Y}^i - \mathbf{Y}^j\right| - \left|\mathbf{X}_{\text{hsi}}^i - \mathbf{X}_{\text{hsi}}^j\right|\right)^2 \tag{21}$$

where $\Omega(i)$ means neighboring regions (top, down, left, and right) centered at the region i and the size of local resign is 4×4. R is the number of local resigns.

The total loss for HR-HSI reconstruction is calculated as

$$L = L_{\text{re}} + \lambda_1 L_{\text{spa}} + \lambda_2 L_{\text{con}} \tag{22}$$

where λ_1 and λ_2 are the hyperparameters to control the contributions.

B. CSMFormer-C for Classification

The proposed CSMFormer for classification (CSMFormer-C) is illustrated in Fig. 5. Intuitively, the LKC is not necessary for land-covers'classification, thus the spatial mixer is achieved by the multiscale strip convolution as shown in Fig. 6.

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The image patches $\mathbf{x} \in \mathbb{R}^{p\times p\times C}$ of reconstructed HR-HSI are fed into the encoder network for feature embedding as

$$\mathbf{x}_e = \mathrm{Conv}_{3\times 3}(\mathbf{x}) \tag{23}$$

and then, the feature embedding is split into two parts as

$$\begin{aligned}\mathbf{x}_e^1 &= \mathrm{norm}(\mathbf{x}_e)[:,:,1:\eta C]\\ \mathbf{x}_e^2 &= \mathrm{norm}(\mathbf{x}_e)[:,:,\eta C:C].\end{aligned} \tag{24}$$

The spatial mixer in CSMA is achieved by the multiscale strip convolution. The combination of horizontal and vertical strip convolutions approximates the standard convolution with fewer parameters. Thus, the long-range spatial–spectral dependencies of the large-scale branch is calculated as

$$\begin{aligned}\mathbf{F}_e^1 &= \mathrm{Conv}_{1\times 1}\left(\mathbf{x}_e^1\right)\\ \mathbf{Att}_s &= \mathrm{DWConv}_3^s\left(\mathbf{F}_e^1\right) + \mathrm{DWConv}_5^s\left(\mathbf{F}_e^1\right)\\ \mathbf{Att} &= \mathrm{Conv}_{1\times 1}(\mathbf{Att}_s)\\ \mathbf{F}_1 &= \mathbf{F}_e^1 \times \mathbf{Att}\end{aligned} \tag{25}$$

where DWConv_k^s denotes a set of strip convolutions $\mathrm{DWConv}_{k\times 1}$ and $\mathrm{DWConv}_{1\times k}$. Finally, the features extracted by the large and original branches are reweighed by (10). After that, the NLEM is utilized to generate the discriminative features $\mathbf{F}_{\mathrm{att}}$, where the calculation is the same as (12).

Finally, a classifier is designed including two fully connected (FC) layers and *softmax* layers. The cross-entropy loss between the real labels and the predicted labels of CSMFormer-C is utilized for optimization

$$L_{\mathrm{ce}} = \sum_{k=1}^{K} -y^k \log p^k \tag{26}$$

where y is the label; if the class is k, then $y^k = 1$, otherwise 0. p^k is the prediction probability after softmax.

III. Experimental Results and Discussion

In this section, two pairs of HSI and MSI are utilized to verify the performance of the proposed CSMFormer baseline. Due to the lack of reference HR-HSI, data fusion evaluation indicators mainly include visual comparison, training metrics, and classification accuracy. The performance of the classification model mainly depends on the classification maps and classification accuracy.

A. Experimental Data

The Yellow River Estuary wetland abounds in tidal flat resources, with typical salt marsh vegetation, such as Spartina alterniflora, Tamarix chinensis, and Suaeda salsa. The HSI and MSI in the Yellow River Estuary scene were taken by ZiYuan1-02D satellite on September 26, 2020, which is equipped with payloads including an advanced hyperspectral imager (AHSI) and multispectral imager. HSI records the spectral information from visible near-infrared (VNIR) to short-wave infrared (SWIR), providing observation with 30-m spatial resolution. For the Yellow River Estuary dataset, the spectral resolution from VNIR is 10 nm with 76 bands, while that of SWIR is 20 nm and 90 bands. The Yellow River Estuary scene comprises 960 × 1170 pixels.

TABLE I
Number of Training and Testing Samples for the Yellow River Estuary Dataset

Yellow River Estuary dataset			
No.	Class	Training	Testing
1	Spartina alterniflora	735	39784
2	Suaeda salsa	2519	118213
3	Tamarix forest	1297	35216
4	Tidal creek	529	15673
5	Mudflat	702	24592
Total		5782	233478

TABLE II
Number of Training and Testing Samples for the Yancheng Dataset

Yancheng dataset			
No.	Class	Training	Testing
1	Sea	30	9474
2	Fallow land	15	1443
3	Salt field	6	633
4	Pond	12	168
5	Rainfed cropland	36	1593
6	Suaeda salsa	22	743
7	Aquaculture pond	16	1298
8	Paddy field	30	4308
9	Estuarine area	36	2061
10	River	42	1497
11	Irrigation canal	18	324
12	Phragmites	24	1164
Total		287	24706

The Yancheng wetland is rich in land resources, which is suitable for the growth of grain, forest, and cash crops. It is one of the areas where land has been growing year after year in China. The Yancheng scene includes HSI and MSI collected by GaoFen-5 and Sentinel-2 satellites, respectively. The AHSI is capable of resolving 330 bands from VNIR to SWIR (400–2500 nm) at 30-m spatial resolution. The spectral resolution of VNIR and SWIR is 5–10 nm, respectively. The bad bands are 1–14, 88, and 129–133 in VNIR, and 42–51, 96–115, 119–123, 171–173, and 175–180 in SWIR. The Sentinel-2 satellite is equipped with a 12-band multispectral camera, which collects land-covers' attributes at 10-m resolution. The Yancheng scene comprises 1800 × 1200 pixels.

The ground-truth (GT) maps for land-covers' classification are annotated through field investigation and visual interpretation. In the training stage, data augmentation is employed to expand training samples, including rotation, flipping, and adding noise. To avoid the overlap between the validation set and the testing set, 20% of the samples are selected from the training set for validation. The pseudo-color images of the multisource remote sensing datasets are shown in Fig. 7. The numbers of training and testing samples for classification are listed in Tables I and II.

B. Evaluation Metrics

For data fusion, the peak signal-to-noise ratio (PSNR) is selected as the primary evaluation metric during the training

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(a) (b)

Fig. 7. Multisource remote sensing datasets. (a) Yellow River Estuary dataset (HSI and MSI). (b) Yancheng dataset (HSI and MSI).

process. In addition, RMSE and the spectral angle mapper (SAM) are also used as auxiliary evaluation metrics. To further verify the advantages of fused images in downstream tasks, the quantitative evaluation of classification is introduced including overall accuracy (OA), average accuracy (AA), and Kappa coefficient (KC).

1) RMSE: RMSE is utilized to evaluate the overall difference between the reconstructed HR-HSI $\mathbf{Y}$ and the reference HSI $\mathbf{X}_hsi$. The RMSE is defined as

$$\mathrm{RMSE} = \sqrt{\frac{\sum_{h=1}^{H}\sum_{w=1}^{W}\sum_{c=1}^{C}\|\mathbf{Y}(h,w,c)-\mathbf{X}_{\mathrm{hsi}}(h,w,c)\|_2}{HWC}}. \tag{27}$$

2) PSNR: PSNR is utilized to evaluate the spatial quality of the reconstructed HR-HSI. The PSNR is defined as

$$\mathrm{PSNR} = \frac{1}{C}\sum_{c=1}^{C} 10\lg\left(\frac{\max(\mathbf{X}_{\mathrm{hsi}}(c))^2}{\frac{1}{HW}\|\mathbf{X}_{\mathrm{hsi}}(c)-\mathbf{Y}(c)\|_2^2}\right) \tag{28}$$

where $\lg(\cdot)$ is a base-10 logarithm. $\mathbf{X}_{\mathrm{hsi}}(c)$ and $\mathbf{Y}(c)$ represent the cth bands of the reconstructed HR-HSI and reference HSI, respectively.

3) SAM: SAM is utilized to evaluate the spectral preservation degree of the reconstructed HR-HSI. The SAM is defined as

$$\mathrm{SAM} = \frac{1}{HW}\sum_{h=1}^{H}\sum_{w=1}^{W}\cos^{-1}\left(\frac{\mathbf{X}_{\mathrm{hsi}}(h,w)^{\mathrm{T}}\mathbf{Y}(h,w)}{\|\mathbf{X}_{\mathrm{hsi}}(h,w)\|\|\mathbf{Y}(h,w)\|}\right) \tag{29}$$

where $\|\cdot\|$ denotes the modular operation of vector. The smaller the SAM value, the higher the spectral preservation degree of the reconstructed HR-HSI.

4) OA and AA: OA and AA are utilized to evaluate the classification precision of the reconstructed HR-HSI. The OA and AA are defined as

$$\mathrm{OA} = \frac{1}{n}\sum_{k=1}^{K} q_k \times 100\% \tag{30}$$

$$\mathrm{AA} = \frac{1}{K}\sum_{k=1}^{K}\frac{q_k}{t_k} \times 100\% \tag{31}$$

where K is the number of classes. q_k represents the number of pixels correctly predicted as class k, and t_k represents the number of real samples of class k.

5) KC: KC is utilized to evaluate the classification consistency of the reconstructed HR-HSI. The KC is defined as

$$\mathrm{KC} = \frac{\mathrm{OA}-P_e}{1-P_e} \tag{32}$$

where

$$P_e = \frac{\sum_{k=1}^{K} t_k \times p_k}{n^2} \tag{33}$$

where p_k denotes the total number of pixels predicted as class k. The higher the KC is, the better the classification consistency is. KC > 0.60 represents substantial consistency, and KC > 0.8 represents almost perfect.

C. Fusion Performance of the CSMFormer-F

To conduct the experimental validation of HS/MS fusion, several state-of-the-art methods are selected for comparison, specifically the Hysure [56], LTMR [12], the Multi-stream Dense Network (MsDNet) [57], SSRNet [20], disentangled representation fusion (DRF) [58], the cross-scale octave network (CSONet) [59], and the LSFormer. Note that the execution of LTMR requires the spatial blurring kernel and spectral response function estimated by HySure. For the HS/MS dataset, the real spectral response functions are unavailable; thus, the mean response is utilized for approximation. The training of deep learning methods depends on the degraded HSI and MSI obtained by resampling. Specifically, each iteration of model training will randomly clip a group of image pairs corresponding to the geographical location from the degraded HSI and MSI. At the same time, an image pair with nonoverlapping positions is used for validation.

1) Analysis of the Preintegrated Method: For HS/MS fusion, three schemes are designed to preintegrate the LR-HSI and HR-MSI, including interpolation, concatenation, and equispaced resampling. The preintegrated method by interpolation is calculated as (2). The preintegrated method by concatenation [see (34)] and equispaced resampling [see (35)] is calculated as

$$\mathbf{X}_{\mathrm{pre}}^{\mathrm{cat}} = \mathrm{cat}\left(\mathrm{Upsampling}\left(\mathbf{X}_{\mathrm{hsi}}^{d}, r\right), \mathbf{X}_{\mathrm{msi}}^{d}\right), \tag{34}$$

$$\mathbf{X}_{\mathrm{pre}}^{\mathrm{resam}}(i) = \begin{cases} \mathrm{Insampling}\left(\mathbf{X}_{\mathrm{msi}}^{d}, r\right)[i], & \text{if } i = \dfrac{b}{3} \\ \mathbf{X}_{\mathrm{hsi}}^{d}[i], & \text{otherwise} \end{cases} \tag{35}$$

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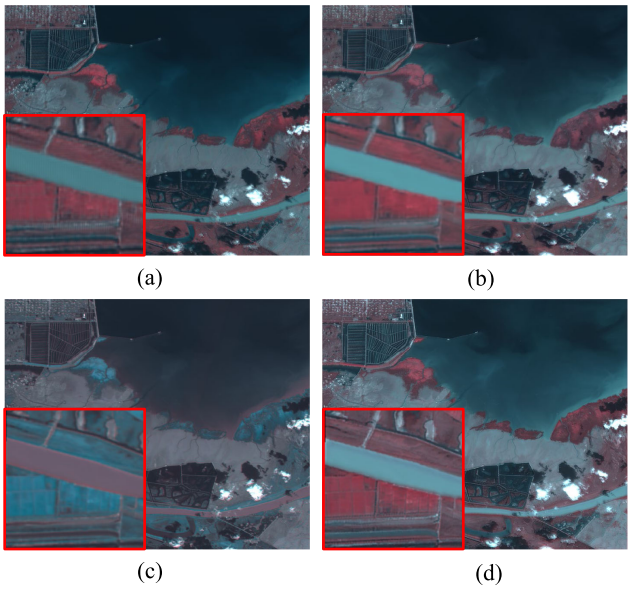


Fig. 8. Fusion results of different preintegrated methods on the Yellow River Estuary dataset. (a) Equispaced resampling. (b) Concatenation. (c) Repeat interpolation. (d) Interpolation.

where cat(·) is the concatenation operator along the channel dimension and Insampling(·) is the equispaced resampling. b is the sampling interval in (2). The equispaced resampling is input with a resolution of 30 m; thus, the upsampling operation is required in the reconstruction process. Inputs of 10-m resolution are utilized for the concatenation and interpolation methods. As shown in Fig. 8(a), the HR-HSI generated by equispaced sampling is relatively blurred, accompanied by blocky artifacts. In contrast, the concatenation and interpolation methods achieve better fusion results. The concatenation ignores the spectral correlation between HSI and MSI, and its fusion effect is lower than the interpolation algorithm. The repeat interpolation increases the density of interpolating spectral bands on the basis of interpolation. Thus, the spectral distortion of reconstructed HR-HSI is large. To sum up, the preintegrated method based on interpolation is superior to the concatenation and equispaced resampling.

2) Analysis of the Hyperparameters: In this article, the hyperparameters λ_1 and λ_2 control the contributions of the gradient loss and consistency loss, respectively. Considering the lack of real reference HR-HSI, only the evaluation metrics at the training stage are reported. Table III reported the relationship between λ and PSNR on the Yellow River estuary dataset. As listed in Table III, the higher PSNR is obtained in the training stage when λ_1 and λ_2 are close to 1.0. When λ_1 and λ_2 are close to 0.01, the proposed model is difficult to focus on reconstructing texture and neighborhood differences, resulting in lower PSNR. Therefore, $\lambda_1 = 1.0$ and $\lambda_2 = 1.0$ are the best choices for the proposed CSMFormer-C.

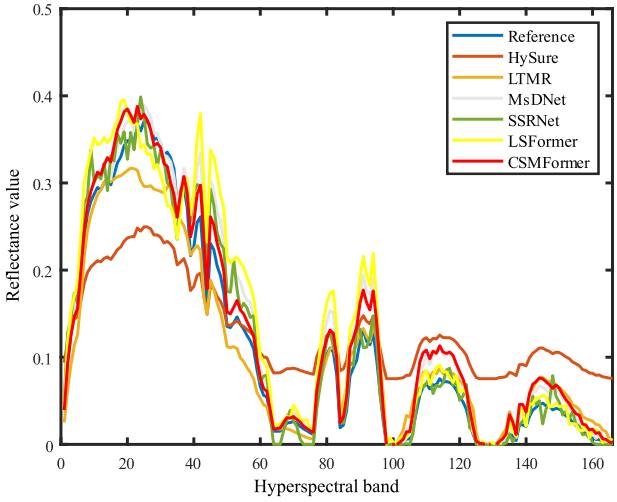
3) Comparison With Related Fusion Methods on the Yellow River Estuary Dataset: Table IV reports the evaluation metrics of the deep learning methods. It is found from Table IV that the DRF and CSONet obtain unsatisfactory training metrics due to the limited number of training samples. The SSRNet has a poor ability to capture long-range dependencies; thus, it does not achieve remarkable performance. The MsDNet benefits from multiscale dense blocks, which achieves promising training performance. In contrast, the proposed CSMFormer-C obtains the consistent superiority of RMSE, SAM, and PSNR on the validation set.

TABLE III
RELATIONSHIP BETWEEN λ AND PSNR ON THE YELLOW RIVER ESTUARY DATASET

Values	$\lambda_1 = 0.01$	$\lambda_1 = 0.1$	$\lambda_1 = 0.5$	$\lambda_1 = 1.0$
$\lambda_2 = 0.01$	41.09	41.13	42.21	42.22
$\lambda_2 = 0.1$	40.81	42.88	42.46	42.10
$\lambda_2 = 0.5$	40.52	42.66	44.65	43.18
$\lambda_2 = 1.0$	40.35	41.36	42.74	**45.21**

TABLE IV
COMPARISON RESULTS OF THE CONSIDERED METHODS ON THE YELLOW RIVER ESTUARY DATASET

Model	Yellow River Estuary		
	RMSE	SAM	PSNR
MsDNet [57]	0.0074	2.45	42.59
SSRNet [20]	0.1480	4.72	36.60
DRF [58]	0.0201	5.74	26.30
CSONet [59]	0.0100	3.02	32.34
LSFormer	0.0077	2.46	42.22
CSMFormer-F	**0.0055**	**2.09**	**45.21**



Method	HySure	LTMR	MsDNet	SSRNet	LSFormer	CSMFormer-F
SAM	0.4226	0.1351	0.1198	0.1191	0.1979	0.1049
M-SAM	0.5071	0.2533	0.1376	0.1535	0.1388	0.1374

Fig. 9. Typical spectral curve of the HR-HSI generated by different methods on the Yellow River Estuary dataset, and the SAM is calculated according to the reference LR-HSI.

Fig. 9 exhibits the typical spectrum of the reconstructed HR-HSIs. Figs. 10 and 11 illustrate some HR-HSIs reconstructed by the competitors with better training performance on the Yellow River Estuary dataset. The 45th, 30th, and 25th bands are selected for exhibiting pseudocolor images. Fig. 10(a) and (b) shows pseudocolor images of LR-HSI and HR-MSI, respectively, and Fig. 11 presents three typical regions of the reconstructed HR-HSI. It is observed from Fig. 10 that the HySure, LTMR, and CSMFormer generate clearer texture details, and the spatial resolution of HR-HSI is significantly improved compared with Fig. 10(a). However, the HySure is obviously difficult to alleviate spectral distortion,

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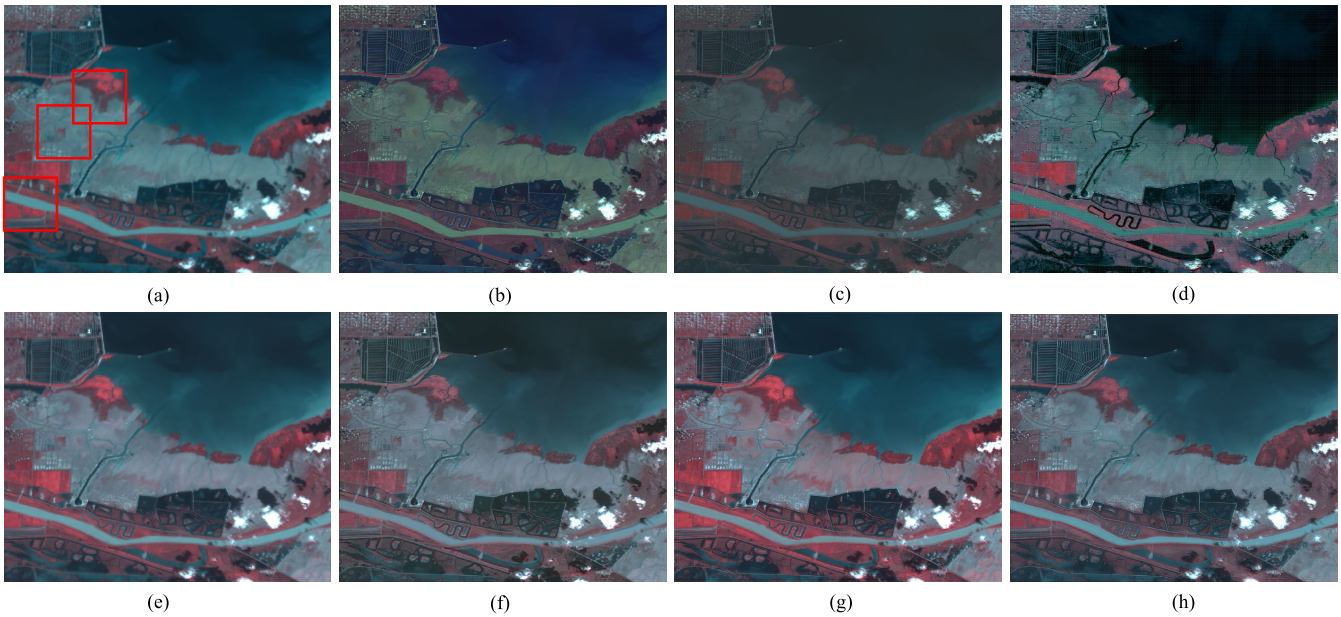


Fig. 10. Fusion results of different methods on the Yellow River Estuary dataset. The red blocks are highlighted in Fig. 11. (a) LR-HSI. (b) HR-MSI. (c) HySure. (d) LTMR. (e) MsDNet. (f) SSRNet. (g) LSFormer. (h) CSMFormer-F.

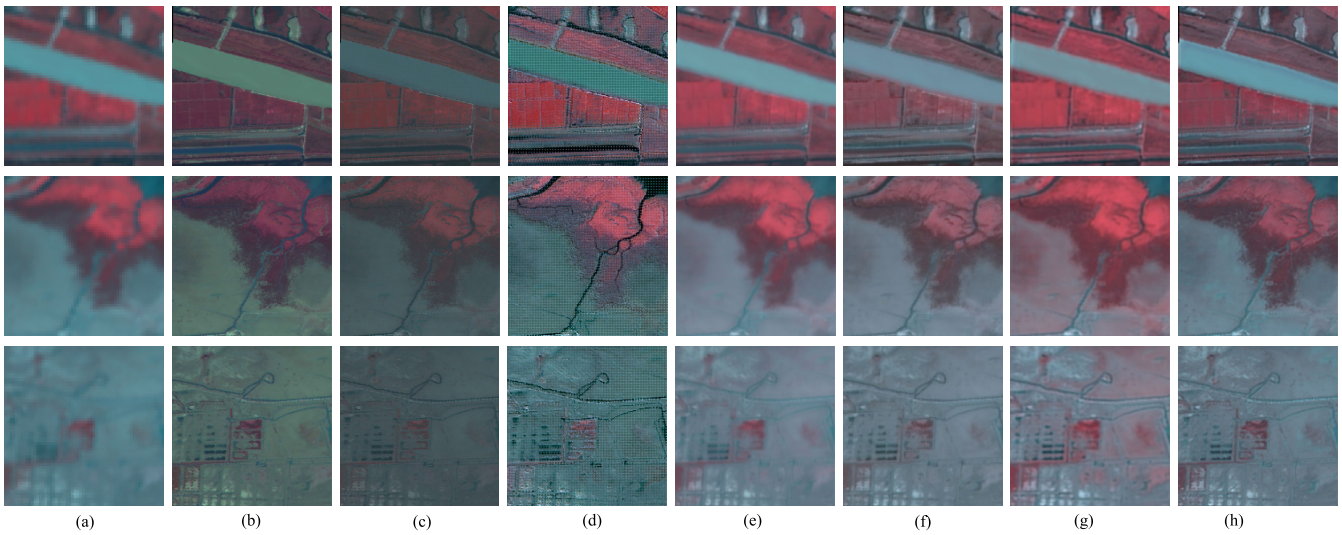


Fig. 11. Local exhibition (red blocks in Fig. 10) of the fusion results on the Yellow River Estuary dataset. (a) LR-HSI. (b) HR-MSI. (c) HySure. (d) LTMR. (e) MsDNet. (f) SSRNet. (g) LSFormer. (h) CSMFormer-F.

while the LTMR generates blocky artifacts. Both of them deteriorate the visualization of the reconstructed HR-HSI. In addition, since only large-scale branches are used in the LSFormer, the fusion performance is also limited by blocky artifacts. In general, the proposed CSMFormer achieves better fusion performance, taking into account texture details and spectral preservation. From Fig. 11, it is further observed that the reconstructed HR-HSI almost all achieve the resolution improvements, which is conducive to visual interpretation. For example, as shown in the second row of Fig. 11, the vegetation boundary (Spartina alterniflora) is separated more clearly by the reconstructed HR-HSI. In the third row of Fig. 11, the LR-HSI can hardly observe the nascent or fragmented vegetation areas, while the HR-HSI has the ability to depict them.

As shown in Fig. 9, a typical spectrum from the reconstructed HR-HSIs is selected for exhibition, including its corresponding SAM and the mean SAM (M-SAM) of the whole images. The downsampled HR-HSIs and the original LR-HSI are utilized for calculating SAM. Considering the lack of a sensor prior, the spectral response function is built by evenly bands correspondence when conducting the HySure. This leads to serious spectral distortion in the fusion map, while the LTMR depends on the spectral response function of HySure. Therefore, a similar trend is also observed in the reported curves and M-SAM. Deep learning methods are more robust in presence of unavailable prior. The proposed CSMFormer-F generally exhibits a lower M-SAM; thus, the spectral distortion is suppressed.

4) Comparison With Related Fusion Methods on the Yancheng Dataset: Table V lists the evaluation metrics of the comparison between the considered deep learning methods and the proposed CSMFormer-F. The DFR and CSONet are difficult to reconstruct complex details with few training

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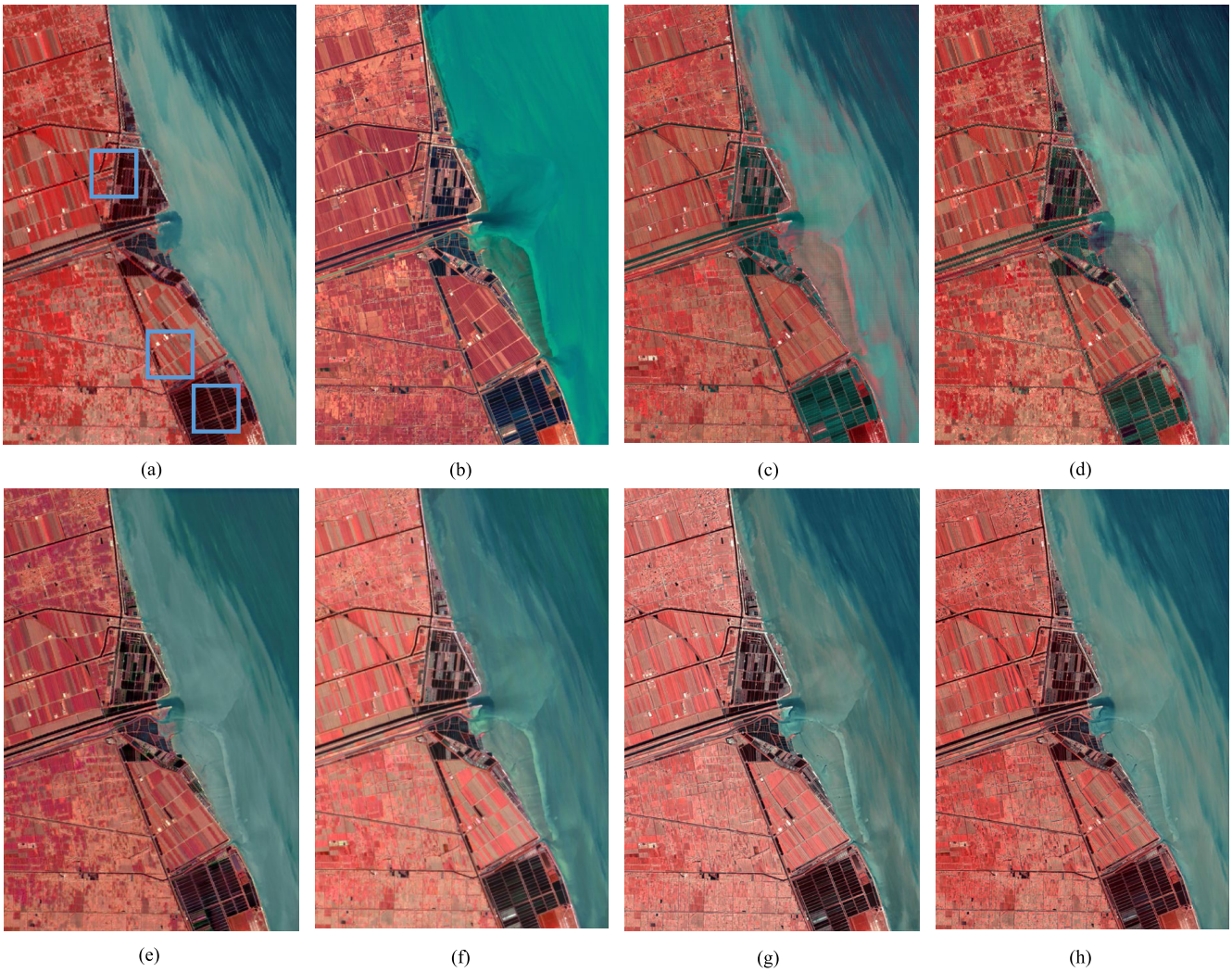


Fig. 12. Fusion results of different methods on the Yancheng dataset. The blue blocks are highlighted in Fig. 13. (a) LR-HSI. (b) HR-MSI. (c) HySure. (d) LTMR. (e) MsDNet. (f) SSRNet. (g) LSFormer. (h) CSMFormer-F.

TABLE V
COMPARISON RESULTS OF THE CONSIDERED METHODS ON THE YANCHENG DATASET

Model	Yancheng		
	RMSE	SAM	PSNR
MsDNet [57]	0.0287	6.13	30.34
SSRNet [20]	0.0275	6.87	30.54
DRF [58]	0.0378	8.58	24.74
CSONet [59]	0.0362	8.77	25.11
LSFormer	0.0192	5.00	31.88
CSMFormer-F	**0.0165**	**4.49**	**33.21**

samples; thus, the PSNR values are relatively low during training. The performance of LSFormer is close to that of the method proposed in this article, but its model complexity is higher. In general, the proposed CSMFormer-F gives consideration to spatial and spectral reconstruction during training, which makes the CSMFormer-F reduce the uncertainty when reconstructing the HR-HSI.

Figs. 12 and 13 exhibit the some reconstructed HR-HSI and several typical areas. The 45th, 30th, and 25th bands are selected for exhibiting pseudocolor images. Because HSI and MSI are collected from different remote sensing satellites, different degrees of spectral distortion affect the visual display of the reconstructed images. The Hysure, LSFormer, and CSMFormer-F reconstruct texture details that are more similar to HR-MSI. The MsDNET and SSRNet also demonstrate competitive detail reconstruction capabilities. It is observed from Fig. 13 that the reconstructed image can effectively alleviate the zigzag boundary problem caused by low resolution. Due to approximately preconstructing the spectral response functions, the HySyre and LTMR obtain excellent spectral reconstruction capability. However, the visualization is deteriorated by blocky artifacts. In contrast, the LSFormer and CSMFormer-F also have the ability to reconstruct spatial texture, and the problem of blocky artifacts has been alleviated. Compared with all competitors, the proposed method is more suitable for HS/MS fusion.

A typical spectrum from the reconstructed HR-HSIs in the Yancheng dataset is exhibited in Fig. 14. Due to the available sensor prior, the HR-HSIs estimated by HySure and LTMR show a more realistic spectrum. It is observed from Fig. 14 that the methods based on deep learning also achieve a competitive

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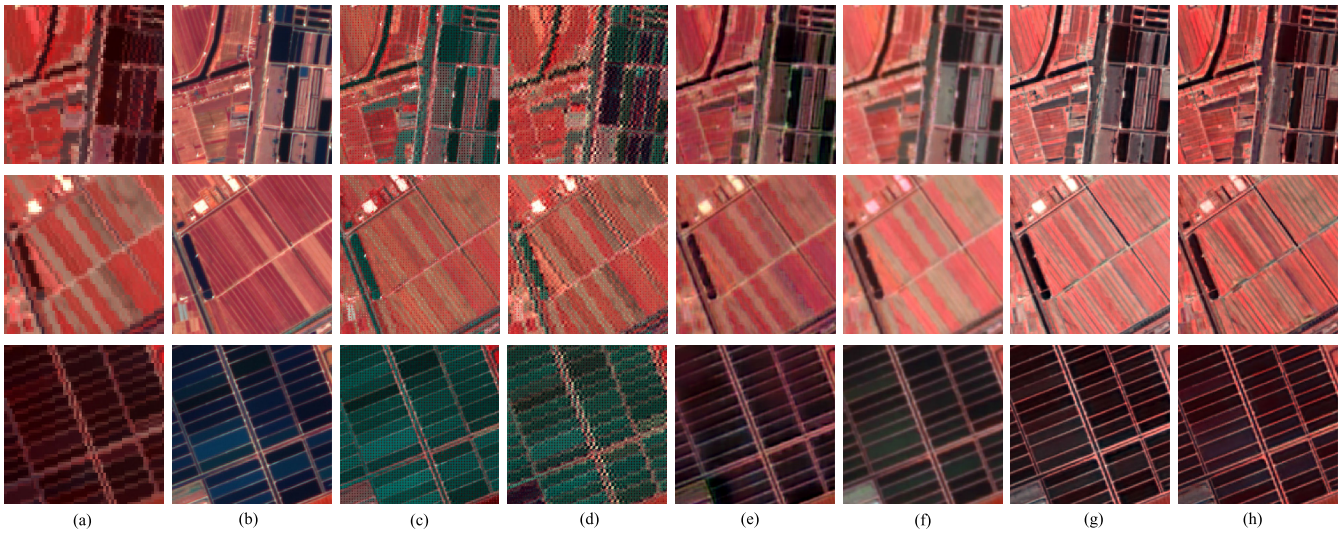
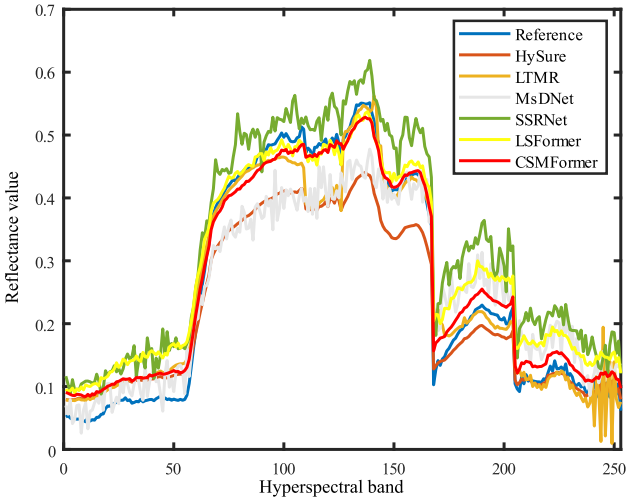


Fig. 13. Local exhibition (blue blocks in Fig. 12) of the fusion results on the Yancheng dataset. (a) LR-HSI. (b) HR-MSI. (c) HySure. (d) LTMR. (e) MsDNet. (f) SSRNet. (g) LSFormer. (h) CSMFormer-F.



Method	HySure	LTMR	MsDNet	SSRNet	LSFormer	CSMFormer-F
SAM	0.2717	0.2907	0.1852	0.1514	0.1442	0.0835
M-SAM	0.0933	0.1157	0.1361	0.1422	0.1246	0.0954

Fig. 14. Typical spectral curve of the HR-HSI generated by different methods on the Yancheng dataset, and the SAM is calculated according to the reference LR-HSI.

performance of spectral reconstruction. The HR-HSI generated by the proposed method has lower spectral distortion without the sensor prior, which is close to the performance of HySure. Although the proposed CSMFormer-F exhibits a certain degree of amplitude shift in the spectral reconstruction, the spectral response is similar to the reference spectrum. Furthermore, the HR-HSI reconstructed by the proposed CSMFormer-F obtains lower SAM and M-SAM, which confirms the excellent ability in spectral reconstruction.

D. Analysis on the Reconstructed HR-HSI by Classification

For the Yellow River estuary dataset, the reconstructed HR-HSIs facilitate the visual interpretation of land-covers. Next, the proposed CSMFormer baseline (CSMFormer-C) is utilized to classify the reconstructed HR-HSIs. It is found from Table VI that, compared with using LR-HSI and HR-MSI alone for classification, the reconstructed HR-HSIs achieve OA improvements. As shown in Fig. 10, the MsDNet suffers from serious noise interference, resulting in blurred spatial texture details. LSFormer presents a serious spectral distortion problem. Therefore, both MsDNet and LSFormer obtain low classification performance. Although the HySure and LTMR suffer from serious blocky artifacts, the reconstruction of texture detail is effective, which encourages better classification performance. In general, the proposed CSMFormer-F shows remarkable performance in HS/HS fusion, which is also conducive to improving classification accuracy.

For the Yancheng dataset, although the classification performance on the reconstructed HR-HSIs is improved, the classification accuracy is subject to varying degrees of spectral distortion. Table VII reports the classification results for the LR-HSI, HR-MSI, and reconstructed HR-HSIs using the proposed CSMFormer-C. In particular, the LTMR obtains the highest OA value among other compared methods, followed by the Hysure. This is because spectral fidelity compensates for the low accuracy caused by blocky artifacts. It is possible to observe that the HR-HSI generated by CSMFormer-F reaches a consistent improvement in the classification performance. This indirectly proves that the proposed CSMFormer-F has the ability to integrate the complementary merits of HSI and MSI, thus improving the classification precision.

In general, the classification precision can reflect the effect of reconstructed HR-HSI to a certain extent. In fact, Tables VI and VII confirm that HR-HSI can significantly improve classification performance. Although the classification precision of HR-HSI under different fusion methods is relatively close, the proposed CSMFormer-F still obtains superior performance. Besides considering the benefits of data fusion for visual interpretation, as shown in Figs. 10 and 12, the proposed CSMFormer-F achieves consistent improvements in terms of spatial–spectral reconstruction and downstream classification.

E. Classification Performance of the CSMFormer-C

To verify the performance of the proposed CSMFormer baseline in classification tasks, several related classifiers

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TABLE VI
CLASSIFICATION RESULTS (%) FOR THE LR-HSI, HR-MSI, AND RECONSTRUCTED HR-HSIS OF YELLOW RIVER ESTUARY DATASET USING THE CSMFORMER-C

No.	LR-HSI	HR-MSI	Hysure [56]	LTMR [12]	MsDNet [57]	SSRNet [20]	LSFormer	CSMFormer-F
1	87.97	84.91	**92.62**	88.57	84.85	90.46	88.21	89.90
2	**92.47**	89.52	80.79	82.40	79.11	85.19	82.58	84.60
3	46.97	51.29	81.88	80.05	68.53	78.55	73.11	**84.44**
4	47.03	54.64	**85.79**	80.98	82.98	79.24	78.26	77.40
5	20.44	30.09	**79.94**	73.61	69.15	71.77	67.94	76.15
OA (%)	74.20	74.36	83.22	82.07	77.70	83.28	80.28	**84.11**
AA (%)	58.97	62.09	84.21	81.12	76.93	81.04	78.02	**82.50**
Kappa	0.5937	0.6072	0.7614	0.7431	0.6816	0.7584	0.7158	**0.7710**

TABLE VII
CLASSIFICATION RESULTS (%) FOR THE LR-HSI, HR-MSI, AND RECONSTRUCTED HR-HSIS OF YANCHENG DATASET USING THE CSMFORMER-C

No.	LR-HSI	HR-MSI	Hysure [56]	LTMR [12]	MsDNet [57]	SSRNet [20]	LSFormer	CSMFormer-F
1	**100.00**	99.74	**100.00**	99.86	**100.00**	99.28	99.61	99.87
2	67.82	37.76	93.86	**95.62**	73.47	98.31	71.84	84.97
3	57.54	93.35	91.73	91.90	99.84	**100.00**	84.12	96.92
4	0.00	20.00	33.33	10.30	20.61	19.39	20.61	**56.36**
5	98.02	67.31	97.70	98.27	97.44	**98.85**	88.16	98.14
6	93.54	40.52	**99.04**	96.15	97.66	89.29	89.97	98.21
7	**99.68**	76.97	91.48	98.34	91.17	82.73	95.98	97.56
8	**99.68**	86.55	99.34	99.03	99.27	99.36	98.98	99.20
9	**100.00**	94.79	**100.00**	**100.00**	99.40	**100.00**	98.86	99.60
10	80.61	89.66	82.72	89.18	89.86	95.99	**91.56**	90.07
11	0.00	74.76	99.68	99.05	95.58	96.21	98.42	**99.68**
12	75.55	96.22	83.47	80.39	**97.89**	78.98	77.48	87.42
OA (%)	92.24	86.52	96.41	96.80	96.25	96.38	94.12	**97.02**
AA (%)	72.67	73.14	89.36	88.18	88.52	88.20	86.63	**92.33**
Kappa	0.9024	0.8309	0.9549	0.9598	0.9530	0.9546	0.9261	**0.9627**

were used for comparison, including spatial–spectral CNN (SSCNN) [60], MiniCAN [45], SpecFormer [51], and SSViT [21].

1) Analysis on the Image Patch Size: Fig. 15 shows the relationship between OA and patch size. It is observed that, as p grows, the OA values grow gradually at the very beginning and change slightly when p reaches 13. The proposed CSMFormer-C improves the classification accuracy when considering a larger patch size. That is, the spatial continuity of the land-covers in the considered datasets is strong. Therefore, the large patch size is beneficial for the proposed to extract the long-range dependencies. In particular, the proposed method obtains the highest OA value when considering 13 $\times$ 13 patch sizes. Too large p hinders the precision because it degenerates the feature discrimination of the center pixel. Therefore, $p = 13$ is suggested in practice to improve classification performance.

2) Comparison With Related Methods on the Yellow River Estuary Dataset: Table VIII and Fig. 16 show the quantitative and qualitative results of the Yellow River Estuary dataset. The Yellow River Estuary scene has seriously mixed vegetation, so all of the models produce noisy results. Specifically, the classification map of SSCNN is fragmented; thus, SSCNN obtains a relatively low OA value. The reason is that SSCNN cannot build long-range dependencies.

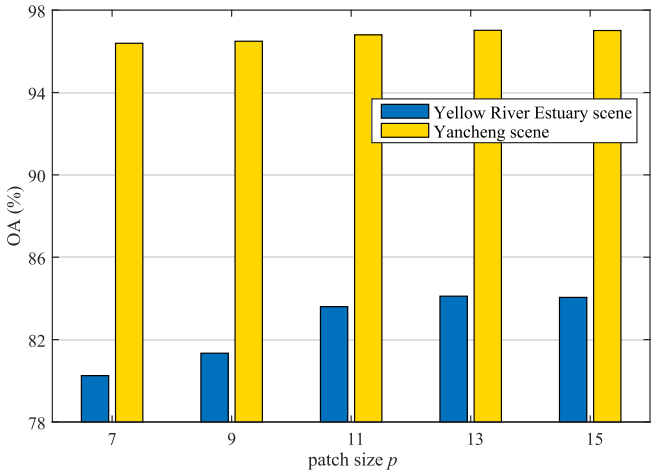


Fig. 15. Relationship between OA and the image patch size.

The proposed CSMFormer-C provides a classification map that is more similar with regard to the corresponding GT map. These results are also consistent with the corresponding evaluation metrics reported in Table VIII. The OA values of the proposed CSMFormer-C are improved by +2.50%, +2.86%, and +1.27% over the MiniCAN, SpecFormer, and

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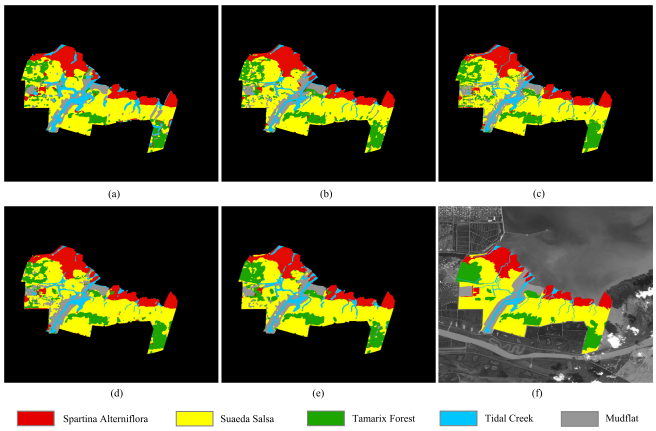


Fig. 16. Full classification maps using different methods on the reconstructed Yellow River Estuary dataset. (a) SSCNN. (b) MiniCAN. (c) SpecFormer. (d) SSViT. (e) CSMFormer-C. (f) Ground truth.

TABLE VIII

CLASS-SPECIFIC CLASSIFICATION ACCURACY (%) USING DIFFERENT METHODS ON THE RECONSTRUCTED YELLOW RIVER ESTUARY DATASET

No.	SSCNN [60]	MiniCAN [45]	SpecFormer [51]	SSViT [21]	CSMFormer-C
1	89.93	92.17	**93.04**	92.86	89.90
2	76.16	82.80	80.55	**84.97**	84.60
3	58.43	74.14	79.16	75.54	**84.44**
4	76.76	74.57	**80.42**	74.28	77.40
5	60.95	73.96	69.05	72.34	**76.15**
OA (%)	74.27	81.61	81.25	82.84	**84.11**
AA (%)	72.45	79.53	80.44	80.00	**82.50**
Kappa	0.6319	0.7332	0.7295	0.7482	**0.7710**

SSViT, respectively. Compared to the proposed CSMFormer-C, the SSViT obtains inferior performance, which indicates that the transformer-based models still have some potential to explore. Similar to the SSViT, the proposed CSMFormer-C constructs long-range spatial–spectral dependencies through the SSMixer; thus, the classification performance is further improved by cross-scale feature representation. In general, the proposed CSMFormer-C achieves consistent superiority in OA values and classification maps.

3) Comparison With Related Methods on the Yancheng Dataset: For illustrative purposes, Fig. 17 presents the full classification maps corresponding to the experiments reported in Table IX. Due to the small number of test sets in the Yancheng dataset, the performance of different classifiers is shown through full classification maps. As it is possible to qualitatively observe in these figures, the classification maps generated by the MiniCAN and SpecFormer tend to be rather noisy. In addition, the SSCNN and MiniCAN hardly identify building areas. The SSViT and proposed CSMFormer-C appear to provide more consistent results in terms of border delineation than the other considered methods. As it is observed in Table IX, all the competitors can achieve over 95% accuracy rate. The SSCNN lacks the ability to mine long-range dependence, while SpecFormer is committed to modeling spectral long-range dependencies, resulting in lower classification accuracy for both SSCNN and SpecFormer. However,

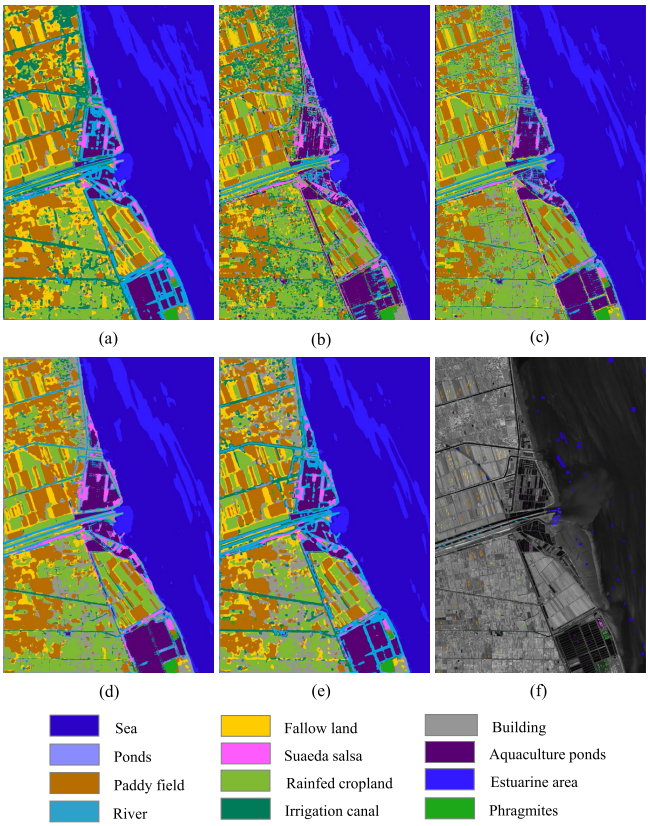


Fig. 17. Full classification maps using different methods on the reconstructed Yancheng dataset. (a) SSCNN. (b) MiniCAN. (c) SpecFormer. (d) SSViT. (e) CSMFormer-C. (f) GT.

TABLE IX

CLASS-SPECIFIC CLASSIFICATION ACCURACY (%) USING DIFFERENT METHODS ON THE RECONSTRUCTED YANCHENG DATASET

No.	SSCNN [60]	MiniCAN [45]	SpecFormer [51]	SSViT [21]	CSMFormer-C
1	99.71	**99.96**	99.96	99.69	99.87
2	84.47	**88.78**	78.55	86.10	84.97
3	96.43	98.70	94.17	**99.35**	96.92
4	41.82	44.85	38.18	38.18	**56.36**
5	90.60	97.44	97.50	96.03	**98.14**
6	89.42	98.21	96.29	97.53	**98.21**
7	95.74	94.24	89.27	**98.58**	97.56
8	98.79	98.63	96.36	**99.31**	99.20
9	**100.00**	98.96	97.97	**100.00**	99.60
10	85.24	92.18	91.36	**98.03**	90.07
11	**99.68**	82.97	97.79	96.21	**99.68**
12	75.37	84.34	79.86	71.06	**87.42**
OA (%)	95.08	96.64	95.02	96.58	**97.02**
AA (%)	88.11	89.94	88.10	90.01	**92.33**
Kappa	0.9382	0.9578	0.9375	0.9570	**0.9627**

the MiniCAN and SSViT achieve competitive classification performance, which proves that spatial–spectral dependencies encourage feature discrimination. In general, the proposed CSMFormer-C is superior to other competitors, and its OA, AA, and Kappa improvements are at least +0.38%, +2.32%, and +0.0049, respectively.

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TABLE X
QUANTITATIVE ANALYSIS OF MODEL COMPLEXITY

Model	Indicators		
	Model Size (MB)	GFLOPs	Running time (s)
MsDNet [57]	167.93	835.67	30.53
SSRNet [20]	2.84	267.84	4.50
LSFormer	12.00	835.67	53.46
CSMFormer-F	6.20	417.21	28.00

F. Analysis of Model Complexity

The quantitative indicators of the proposed CSMFormer baseline and other deep learning methods are reported in Table X. Taking CSMFormer-F in the Yellow River Estuary scene as an example, all experiments are implemented on the PyTorch 1.9.0 software platform. The hardware environment consists of a CPU (AMD Ryzen 75800X 8-Core), 40 GB of DDR4, and Nvidia GTX 3070Ti GPU. Considering the limitation of the hardware environment, the input size is $1 \times 166 \times 600 \times 600$ when calculating giga floating point operations (GFLOPs). It is observed from Table X that, although the model complexity of SSRNet is low, its ability to extract long-range dependences is poor. The proposed CSMFormer-F obtains suboptimal model complexity, which is about half of that of LSFormer. This is because the two-branch strategy is implemented in CSMFormer-F to obtain both long-range and local dependencies. The parameter quantity is reduced to some extent. Concretely, as shown in Fig. 4, the parameter quantity of two branches in CSMFormer-F is $(C^2/2) \times 4 + 5 \times 5 \times (C/2) + 7 \times 7 \times (C/2) = 2C^2 + 37C$, while the parameter quantity of LSFormer is $C^2 \times 3 + 5 \times 5 \times C + 7 \times 7 \times C = 3C^2 + 74C$. That is to say, compared with the LSFormer, which only employs the large-scale branch, the model size of CSMFormer-F is reduced by nearly half. Similarly, for the structural expansion of other spatial mixers, as shown in Fig. 6, CSMFormer-F also has lower model complexity. Although the MsDNet is complex, the performance test time is competitive. This is because the depthwise convolution of LSFormer and CSMFormer actually increases the time cost due to the grouping operation. In general, the proposed CSMFormer has achieved remarkable results in terms of complexity and performance tradeoffs.

IV. CONCLUSION

A transformer-based baseline was proposed for HS/MS fusion classification in real wetland scenes. The proposed CSMFormer baseline benefits from two aspects. One is that SSMixers extract long-range dependencies, thus improving the ability of feature representation. The SSMixer of the proposed CSMFormer baseline can be replaced to adapt to different tasks, including fusion, segmentation, and classification. Specifically, the spatial mixer with larger receptive fields is employed for HS/MS fusion, and the convolutional kernel decomposition strategy is utilized to reduce computational complexity and improve the applicability. For classification tasks, the small receptive fields are more suitable for spatial mixers. Therefore, the multiscale strip convolution is employed to extract long-range spatial dependences. The other one is that cross-scale attention is developed for feature calibration from the original scale to a larger scale, and feature splits for cross-scale reduce the computational burden. Specifically, the partial features are used for feature extraction at the original scale, while SSMixers are implemented on the remaining features at the larger scale. Cross-scale feature calibration can encourage feature discrimination. Extensive experiments have been conducted for HS/MS fusion classification in the Yellow River Estuary and Yancheng scenes, which demonstrates that the reconstructed HR-HSIs improve the visual interpretation and classification performance. Through experimental comparison, the proposed CSMFormer baseline is superior to other competitors in both HS/MS fusion and classification tasks. However, the proposed method still has some limitations in spatial texture reconstruction. In the future, we will conduct extensive tests analyzing the spatial–spectral reconstruction tradeoff of our newly proposed method by different loss functions.

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